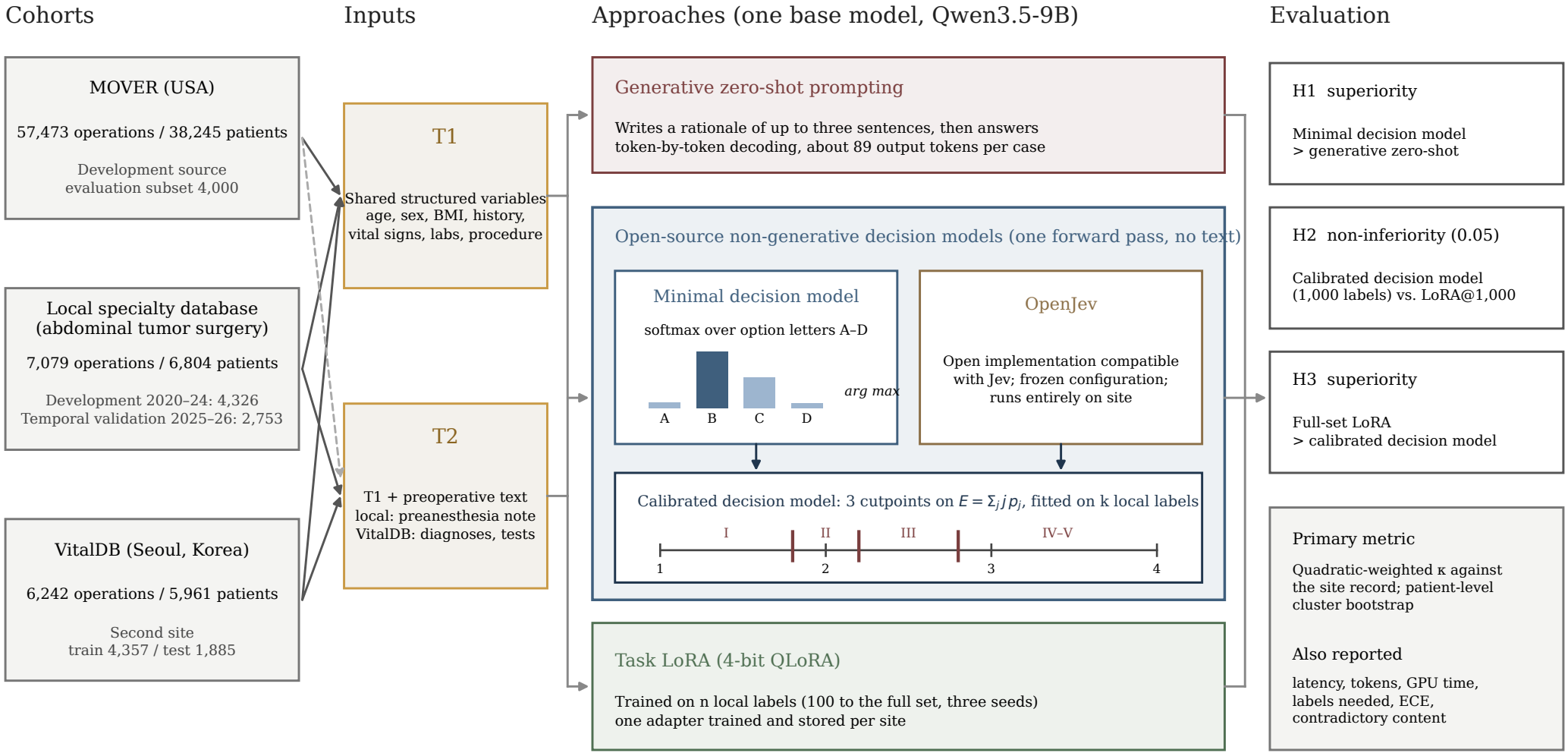
Cohorts
Inputs
Approaches (one base model, Qwen3.5-9B)
Evaluation
MOVER (USA)
57,473 operations / 38,245 patients
Development source
evaluation subset 4,000
Local specialty database
(abdominal tumor surgery)
7,079 operations / 6,804 patients
Development 2020–24: 4,326
Temporal validation 2025–26: 2,753
VitalDB (Seoul, Korea)
6,242 operations / 5,961 patients
Second site
train 4,357 / test 1,885
T1
Shared structured variables
age, sex, BMI, history,
vital signs, labs, procedure
T2
T1 + preoperative text
local: preanesthesia note
VitalDB: diagnoses, tests
Generative zero-shot prompting
Writes a rationale of up to three sentences, then answers
token-by-token decoding, about 89 output tokens per case
Open-source non-generative decision models (one forward pass, no text)
Minimal decision model
softmax over option letters A–D
arg max
A
B
C
D
OpenJev
Open implementation compatible
with Jev; frozen configuration;
runs entirely on site
Calibrated decision model: 3 cutpoints on $E = \Sigma_j j\, p_j$, fitted on k local labels
I
II
III
IV–V
1
2
3
4
Task LoRA (4-bit QLoRA)
Trained on n local labels (100 to the full set, three seeds)
one adapter trained and stored per site
H1 superiority
Minimal decision model
> generative zero-shot
H2 non-inferiority (0.05)
Calibrated decision model
(1,000 labels) vs. LoRA@1,000
H3 superiority
Full-set LoRA
> calibrated decision model
Primary metric
Quadratic-weighted κ against
the site record; patient-level
cluster bootstrap
Also reported
latency, tokens, GPU time,
labels needed, ECE,
contradictory content